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Abbreviation: AM: aromatherapy massage; AT: acupoint therapy; C: control, CD: cool dialysate; DA: dopamine agonist; G: gabapentin; IAE: intradialytic aerobic exercise; ISE: intradialytic stretching exercise; NES: neuromuscular electrical stimulation; Vit-C: vitamin C; Vit-E: vitamin E; R: reflexology

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Supplemental Figure 8. Assessment of risk of bias of including studies.

Supplemental Document

Assessment risk of bias of included trials

Some sources of potential bias included the following: (1) seven enrolled RCTs had some concern of bias related to domain 1 due to no detail information of allocation; one RCT (Deng, 2017) had high risk of bias related to domain 1 due to not a blinded study and participants could have been aware of the nature of the medication taken. (ii) three enrolled RCTs had high risk of bias related to domain 3, two of them (Hashemi, 2015, Mohammadi, 2018) had relatively high rate of loss follow up but not reported the reason and another one RCT (Dauvilliers, 2016) reported that patients dropped out due to feel less efficacy of intervention. (iii) seven enrolled RCTs had a high risk of bias related to domain 4, five of them were not double blinded design studies and two of them did not report the detailed information that participants were aware of the interventions or not. Most of the studies that did not double blinded design were the studies with intradialytic exercise as one of the interventions which made the blinded design become challenged. The outcome measurement methods used in all of the included trials had no diagnostic detection bias, and there were no studies that had multiple eligible outcome measurements within the outcome domain, yielding the low risk in measurement outcomes and results reported.

Grading the evidence of the network meta-analysis using CINeMA

1. Within-study bias: As above mentioned, 12 of enrolled studies were low risk of bias, 2 of enrolled studies were moderate risk of bias and 10 of enrolled studies were high risk of bias.
2. Reporting bias: Our search included gray literature and we also checked unpublished study in ClinicalTrials.gov using keywords: “dialysis OR end stage renal disease” AND “restless legs syndrome”. Total 15 relevant trials identified in ClinicalTrials.gov. Only 1 completed studies are not published yet with remaining others are under recruiting or completed but without different outcome of interest. No small study bias was observed by our funnel plot and Egger’s test, also. Overall, we considered no concern in this domain.
3. Indirectness: The indirectness in NMA is judged according to relevance of enrolled studies to the research question. We thought all enrolled studies to be no concern in this domain.
4. Imprecision: We defined clinically important size of effect with mean difference of RLS score at least 3. In comparison to the control group, the treatment effect of 8 interventions (cool dialysate, aromatherapy massage, intradialytic stretching exercise, reflexology, vitamin C, gabapentin, dopamine agonist, vitamin C + vitamin E) were

considered no concern , two interventions (iron,vitamin E) were some concern and four interventions (acupoint therapy,intradialytic aerobic exercise, intradialytic aerobic exercise + dopamine agonist, neuromuscular electrical stimulation) were considered as major concern regarding imprecision.

5. Heterogeneity: The prediction intervals in relation to clinically important effect size were used to judge the variation in treatment effects between enrolled studies. In comparison to the control group, the treatment effect of seven interventions (cool dialysate, aromatherapy massage, acupoint therapy, intradialytic aerobic exercise, intradialytic stretching exercise, intradialytic aerobic exercise + dopamine agonist, neuromuscular electrical stimulation) were considered no concern , six interventions (dopamine agonist, reflexology, vitamin C, iron, vitamin E, gabapentin) were some concern and one interventions (vitamin C + vitamin E) were considered as major concern regarding heterogeneity.

6. Incoherence: The global approach for incoherence using the design-by-treatment interaction model was conducted with Q value: 22.19 and p value : 0.001. The local approach was conducted using the Separate Indirect from Direct Evidence (SIDE and node-splitting approach) method. Loop inconsistency were not detected with the p value of differences from direct & indirect mean differences were all more than 0.1. For comparisons that had only indirect evidence available, we downgraded the incoherence to major concern levels owing to the p-value of the design by treatment interaction model was less than 0.01.

We summarized the judgments across domains using the 4 levels of confidence according to the GRADE approach: very low, low, moderate, or high. We started the rating from a high level of confidence and we dropped the level of confidence by 1 for each domain with some concerns and dropped the level of confidence by 2 for each domain with major concerns. Furthermore, for interrelated domains (imprecision, heterogeneity, and incoherence), downgrading by a total 2 levels was considered maximum. In summary, only one of 14 interventions was rated as high level of confidence (cool dialysate) and two of 14 interventions were with moderate confidence (vitamin C, vitamin E). Five were with low confidence (acupoint therapy, aromatherapy massage, intradialytic stretching exercise, Vitamin C+Vitamin E, intradialytic aerobic exercise+DA) and the remaining 6 were with very low confidence (**Table S7**).

Supplemental Table 1. PRISMA extension checklist of current network meta-analysis

Section/Topic	Item	Checklist Item	Page
TITLE			
Title	1	Identify the report as a systematic review *incorporating a network meta-analysis (or related form of meta-analysis)*.	1
ABSTRACT Structured summary	2	Provide a structured summary including, as applicable: **Background:** main objectives **Methods:** data sources; study eligibility criteria, participants, and interventions; study appraisal; and *synthesis methods, such as network meta-analysis.* **Results:** number of studies and participants identified; summary estimates with corresponding confidence/credible intervals; *treatment rankings may also be discussed. Authors may choose to summarize pairwise comparisons against a chosen treatment included in their analyses for brevity.* **Discussion/Conclusions:** limitations; conclusions and implications of findings. **Other:** primary source of funding; systematic review registration number with registry name.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known*, including mention of why a network meta-analysis has been conducted.*	5
Objectives	4	Provide an explicit statement of questions being addressed, with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate whether a review protocol exists and if and where it can be accessed (e.g., Web address); and, if available, provide registration information, including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. *Clearly describe eligible treatments included in the treatment network, and note whether any have been clustered or merged into the same node (with justification).*	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6,
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6, Table S2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7

Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Geometry of the network	**S1**	Describe methods used to explore the geometry of the treatment network under study and potential biases related to it. This should include how the evidence base has been graphically summarized for presentation, and what characteristics were compiled and used to describe the evidence base to readers.	8
Risk of bias within individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means). *Also describe the use of additional summary measures assessed, such as treatment rankings and surface under the cumulative ranking curve (SUCRA) values, as well as modified approaches used to present summary findings from meta-analyses.*	8
Planned methods of analysis	14	Describe the methods of handling data and combining results of studies for each network meta-analysis. This should include, but not be limited to: • *Handling of multi-arm trials;* • *Selection of variance structure;* • *Selection of prior distributions in Bayesian analyses; and* • *Assessment of model fit.*	8
Assessment of Inconsistency	**S2**	Describe the statistical methods used to evaluate the agreement of direct and indirect evidence in the treatment network(s) studied. Describe efforts taken to address its presence when found.	8-9
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9
Additional analyses	16	Describe methods of additional analyses if done, indicating which were pre-specified. This may include, but not be limited to, the following: • Sensitivity or subgroup analyses; • Meta-regression analyses; • *Alternative formulations of the treatment network; and* • *Use of alternative prior distributions for Bayesian analyses (if applicable).*	9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9 Fig S1
Presentation of network structure	**S3**	Provide a network graph of the included studies to enable visualization of the geometry of the treatment network.	9 Fig 1
Summary of network geometry	**S4**	Provide a brief overview of characteristics of the treatment network. This may include commentary on the abundance of trials and randomized patients for the different interventions and pairwise comparisons in the network, gaps of evidence in the treatment network, and potential biases reflected by the network structure.	9-10

Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment.	14
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: 1) simple summary data for each intervention group, and 2) effect estimates and confidence intervals. *Modified approaches may be needed to deal with information from larger networks.*	Tab S3-5 Tab 2
Synthesis of results	21	Present results of each meta-analysis done, including confidence/credible intervals. *In larger networks, authors may focus on comparisons versus a particular comparator (e.g. placebo or standard care), with full findings presented in an appendix. League tables and forest plots may be considered to summarize pairwise comparisons.* If additional summary measures were explored (such as treatment rankings), these should also be presented.	11-13 Fig 2-3
Exploration for inconsistency	**S5**	Describe results from investigations of inconsistency. This may include such information as measures of model fit to compare consistency and inconsistency models, *P* values from statistical tests, or summary of inconsistency estimates from different parts of the treatment network.	11-14
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies for the evidence base being studied.	13-14
Results of additional analyses	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression analyses*, alternative network geometries studied, alternative choice of prior distributions for Bayesian analyses,* and so forth).	13
DISCUSSION			
Summary of evidence	24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy-makers).	14, Fig 4 Tab 2
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias). *Comment on the validity of the assumptions, such as transitivity and consistency. Comment on any concerns regarding network geometry (e.g., avoidance of certain comparisons).*	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	18
FUNDING Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. This should also include information regarding whether funding has been received from manufacturers of treatments in the network and/or whether some of the authors are content experts with professional conflicts of interest that could affect use of treatments in the network.	21

Supplemental Table S2. Search strategy for each database

No language restrictions were applied.

Pubmed through May 01st, 2021

#1	(Restless Legs Syndrome[Title/Abstract]) OR (Restless Legs Syndrome[MeSH Terms]) OR (Restless Leg Syndrome[Title/Abstract])
#2	((((((dialysis[Title/Abstract]) OR (Chronic kidney failure[Title/Abstract])) OR (Chronic renal failure[Title/Abstract])) OR (End stage renal disease[Title/Abstract])) OR (End stage kidney disease[Title/Abstract])) OR (Kidney Failure, Chronic[MeSH Terms])) OR (Renal Dialysis[MeSH Terms])
#3	#1 AND #2

EMbase through May 01st, 2021

#1	'restless legs syndrome'/exp OR 'restless legs syndrome'
#2	'end stage renal disease'/exp OR 'end stage renal disease'
#3	'dialysis'/exp OR dialysis
#4	'hemodialysis'/exp OR hemodialysis
#5	'peritoneal dialysis'/exp OR 'peritoneal dialysis'
#6	'chronic kidney failure'/exp OR 'chronic kidney failure'
#7	#2 OR #3 OR #4 OR #5 OR #6
#8	#1 AND #7

Medline through May 01st, 2021

#1	restless legs syndrome.mp. or Restless Legs Syndrome/
#2	end stage renal disease.mp. or Kidney Failure, Chronic/
#3	hemodialysis.mp. or Renal Dialysis/
#4	peritoneal dialysis.mp. or Peritoneal Dialysis/
#5	'renal replacement therapy'/exp

#6	chronic renal failure.mp. or Kidney Failure, Chronic/
#7	2 or 3 or 4 or 5
#8	1 and 6

Supplemental Table 3. League table of effect size of different interventions for restless legs syndrome symptom relief

CD	1.00 (-7.43 to 9.43)	.	.	.	.	.	.	.	.	.	.	.	.	20.14 (12.28 to 28.00)
4.82 (-1.51 to 11.15)	ISE	.	.	1.34 (-6.72 to 9.40)	.	.	.	.	.	.	.	.	.	12.04 (6.34 to 17.74)
5.92 (-1.42 to 13.26)	1.10 (-5.24 to 7.44)	AM	.	.	.	.	.	.	.	.	.	.	.	10.91 (6.96 to 14.85)
7.93 (-1.38 to 17.24)	3.11 (-5.43 to 11.64)	2.01 (-5.98 to 10.00)	G	.	.	.	.	1.75 (-2.93 to 6.44)	.	.	.	.	.	.
8.77 (1.00 to 16.55)	3.95 (-2.34 to 10.24)	2.86 (-3.76 to 9.47)	0.85 (-7.90 to 9.60)	R	.	.	.	.	.	.	.	.	.	8.50 (2.97 to 14.03)
9.09 (0.75 to 17.44)	4.27 (-3.20 to 11.75)	3.18 (-3.66 to 10.01)	1.17 (-6.94 to 9.28)	0.32 (-7.39 to 8.03)	Vit-C	-0.30 (-8.36 to 7.76)	-0.10 (-8.29 to 8.09)	-0.59 (-9.30 to 8.12)	.	.	.	.	.	8.51 (2.69 to 14.34)
9.21 (-0.56 to 18.99)	4.39 (-4.65 to 13.44)	3.30 (-5.23 to 11.82)	1.29 (-8.64 to 11.22)	0.44 (-8.80 to 9.69)	0.12 (-7.47 to 7.71)	Vit-C+ Vit-E	0.20 (-8.23 to 8.63)	.	.	.	.	.	.	7.20 (-0.81 to 15.21)
9.41 (-0.46 to 19.29)	4.59 (-4.56 to 13.75)	3.50 (-5.15 to 12.14)	1.49 (-8.55 to 11.52)	0.64 (-8.71 to 9.99)	0.32 (-7.40 to 8.04)	0.20 (-8.23 to 8.63)	Vit-E	.	.	.	.	.	.	7.00 (-1.14 to 15.14)
9.68 (1.64 to 17.72)	4.86 (-2.28 to 12.00)	3.76 (-2.71 to 10.23)	1.75 (-2.93 to 6.44)	0.91 (-6.48 to 8.30)	0.59 (-6.03 to 7.20)	0.47 (-8.29 to 9.22)	0.27 (-8.61 to 9.14)	DA	.	.	0.85 (-9.49 to 11.19)	.	.	7.34 (2.05 to 12.63)
10.25 (0.47 to 20.04)	5.43 (-3.62 to 14.49)	4.34 (-4.20 to 12.87)	2.33 (-7.95 to 12.61)	1.48 (-7.77 to 10.73)	1.16 (-8.25 to 10.57)	1.04 (-9.66 to 11.74)	0.84 (-9.96 to 11.63)	0.57 (-8.57 to 9.72)	Iron	.	.	.	.	6.57 (-1.00 to 14.14)
11.70 (-4.56 to 27.96)	6.88 (-8.95 to 22.71)	5.78 (-9.76 to 21.32)	3.77 (-12.23 to 19.77)	2.92 (-13.02 to 18.87)	2.60 (-13.28 to 18.49)	2.48 (-14.27 to 19.24)	2.28 (-14.53 to 19.10)	2.02 (-13.28 to 17.32)	1.44 (-15.39 to 18.28)	IAE+DA	1.30 (-10.96 to 13.56)	.	.	.
13.00 (2.32 to 23.68)	8.18 (-1.84 to 18.19)	7.08 (-2.47 to 16.63)	5.07 (-5.21 to 15.35)	4.22 (-5.97 to 14.42)	3.90 (-6.20 to 14.01)	3.78 (-7.64 to 15.21)	3.58 (-7.93 to 15.10)	3.32 (-5.83 to 12.47)	2.74 (-8.79 to 14.28)	1.30 (-10.96 to 13.56)	IAE	.	4.80 (-2.64 to 12.24)	2.51 (-6.56 to 11.58)
15.22 (7.08 to 23.35)	10.40 (3.16 to 17.64)	9.30 (2.72 to 15.88)	7.29 (-1.43 to 16.02)	6.44 (-1.04 to 13.93)	6.12 (-1.56 to 13.81)	6.00 (-3.22 to 15.22)	5.80 (-3.52 to 15.13)	5.54 (-1.82 to 12.90)	4.96 (-4.26 to 14.19)	3.52 (-12.41 to 19.45)	2.22 (-7.95 to 12.39)	AT	.	1.61 (-3.67 to 6.88)
17.80 (4.78 to 30.81)	12.98 (0.50 to 25.45)	11.88 (-0.23 to 23.99)	9.87 (-2.82 to 22.56)	9.02 (-3.60 to 21.65)	8.70 (-3.84 to 21.25)	8.58 (-5.05 to 22.22)	8.38 (-5.32 to 22.09)	8.12 (-3.67 to 19.91)	7.54 (-6.18 to 21.27)	6.10 (-8.24 to 20.44)	4.80 (-2.64 to 12.24)	2.58 (-10.02 to 15.18)	NES	.
16.82 (10.63 to 23.02)	12.00 (7.04 to 16.97)	10.91 (6.96 to 14.85)	8.90 (1.95 to 15.85)	8.05 (2.73 to 13.37)	7.73 (2.14 to 13.32)	7.61 (0.05 to 15.17)	7.41 (-0.28 to 15.10)	7.14 (2.01 to 12.27)	6.57 (-1.00 to 14.14)	5.13 (-9.91 to 20.16)	3.83 (-4.87 to 12.53)	1.61 (-3.67 to 6.88)	-0.97 (-12.42 to 10.47)	Control

Intervention effects are provided as mean difference and 95% CI). Mean difference more than 3 indicate a clinical significant effect for the

column-defining intervention for the network meta-analysis results (lower triangle) and the row-defining intervention for the pair-wise meta-analysis results (upper triangle)

Abbreviation: AM: aromatherapy massage; AT: acupoint therapy; C: control, CD: cool dialysate; DA: dopamine agonist; G: gabapentin; IAE: intradialytic aerobic exercise; ISE: intradialytic stretching exercise; NES: neuromuscular electrical stimulation; Vit-C: vitamin C; Vit-E: vitamin E; R: reflexology

Supplemental Table 4. League table of effect size of different interventions for sleep quality improvement

Gabapentin	.	1.48 (0.72 to 2.23)	.	.
0.66 (-1.03 to 2.35)	Acupoint therapy	.	1.34 (0.62 to 2.06)	.
1.48 (0.72 to 2.23)	0.82 (-0.70 to 2.33)	DA	0.55 (-0.95 to 2.05)	0.61 (-0.69 to 1.91)
2.00 (0.47 to 3.53)	1.34 (0.62 to 2.06)	0.52 (-0.80 to 1.85)	Control	0.10 (-0.64 to 0.84)
2.10 (0.64 to 3.56)	1.44 (0.40 to 2.48)	0.62 (-0.63 to 1.88)	0.10 (-0.64 to 0.84)	Exercise

Intervention effects are provided as standardized mean difference and 95% CI. Standardized mean difference more than 0 indicate a potential benefit effect for sleep quality improvement for the network meta-analysis results (lower triangle) and the row-defining intervention for the pair-wise meta-analysis results (upper triangle)

Abbreviation: DA: dopamine agonist

Supplemental Table 5. League table of treatments-related adverse events

CD	1.00 (-7.43 to 9.43)	.	.	.	.	.	.	.	.	.	.	.	.	20.14 (12.28 to 28.00)
4.82 (-1.51 to 11.15)	ISE	.	.	1.34 (-6.72 to 9.40)	.	.	.	.	.	.	.	.	.	12.04 (6.34 to 17.74)
5.92 (-1.42 to 13.26)	1.10 (-5.24 to 7.44)	AM	.	.	.	.	.	.	.	.	.	.	.	10.91 (6.96 to 14.85)
7.93 (-1.38 to 17.24)	3.11 (-5.43 to 11.64)	2.01 (-5.98 to 10.00)	G	.	.	.	.	1.75 (-2.93 to 6.44)	.	.	.	.	.	.
8.77 (1.00 to 16.55)	3.95 (-2.34 to 10.24)	2.86 (-3.76 to 9.47)	0.85 (-7.90 to 9.60)	R	.	.	.	.	.	.	.	.	.	8.50 (2.97 to 14.03)
9.09 (0.75 to 17.44)	4.27 (-3.20 to 11.75)	3.18 (-3.66 to 10.01)	1.17 (-6.94 to 9.28)	0.32 (-7.39 to 8.03)	Vit-C	-0.30 (-8.36 to 7.76)	-0.10 (-8.29 to 8.09)	-0.59 (-9.30 to 8.12)	.	.	.	.	.	8.51 (2.69 to 14.34)
9.21 (-0.56 to 18.99)	4.39 (-4.65 to 13.44)	3.30 (-5.23 to 11.82)	1.29 (-8.64 to 11.22)	0.44 (-8.80 to 9.69)	0.12 (-7.47 to 7.71)	Vit-C + Vit-E	0.20 (-8.23 to 8.63)	.	.	.	.	.	.	7.20 (-0.81 to 15.21)
9.41 (-0.46 to 19.29)	4.59 (-4.56 to 13.75)	3.50 (-5.15 to 12.14)	1.49 (-8.55 to 11.52)	0.64 (-8.71 to 9.99)	0.32 (-7.40 to 8.04)	0.20 (-8.23 to 8.63)	Vit-E	.	.	.	.	.	.	7.00 (-1.14 to 15.14)
9.68 (1.64 to 17.72)	4.86 (-2.28 to 12.00)	3.76 (-2.71 to 10.23)	1.75 (-2.93 to 6.44)	0.91 (-6.48 to 8.30)	0.59 (-6.03 to 7.20)	0.47 (-8.29 to 9.22)	0.27 (-8.61 to 9.14)	DA	.	.	0.85 (-9.49 to 11.19)	.	.	7.34 (2.05 to 12.63)
10.25 (0.47 to 20.04)	5.43 (-3.62 to 14.49)	4.34 (-4.20 to 12.87)	2.33 (-7.95 to 12.61)	1.48 (-7.77 to 10.73)	1.16 (-8.25 to 10.57)	1.04 (-9.66 to 11.74)	0.84 (-9.96 to 11.63)	0.57 (-8.57 to 9.72)	Iron	.	.	.	.	6.57 (-1.00 to 14.14)
11.70 (-4.56 to 27.96)	6.88 (-8.95 to 22.71)	5.78 (-9.76 to 21.32)	3.77 (-12.23 to 19.77)	2.92 (-13.02 to 18.87)	2.60 (-13.28 to 18.49)	2.48 (-14.27 to 19.24)	2.28 (-14.53 to 19.10)	2.02 (-13.28 to 17.32)	1.44 (-15.39 to 18.28)	IAE+DA	1.30 (-10.96 to 13.56)	.	.	.
13.00 (2.32 to 23.68)	8.18 (-1.84 to 18.19)	7.08 (-2.47 to 16.63)	5.07 (-5.21 to 15.35)	4.22 (-5.97 to 14.42)	3.90 (-6.20 to 14.01)	3.78 (-7.64 to 15.21)	3.58 (-7.93 to 15.10)	3.32 (-5.83 to 12.47)	2.74 (-8.79 to 14.28)	1.30 (-10.96 to 13.56)	IAE	.	4.80 (-2.64 to 12.24)	2.51 (-6.56 to 11.58)
15.22 (7.08 to 23.35)	10.40 (3.16 to 17.64)	9.30 (2.72 to 15.88)	7.29 (-1.43 to 16.02)	6.44 (-1.04 to 13.93)	6.12 (-1.56 to 13.81)	6.00 (-3.22 to 15.22)	5.80 (-3.52 to 15.13)	5.54 (-1.82 to 12.90)	4.96 (-4.26 to 14.19)	3.52 (-12.41 to 19.45)	2.22 (-7.95 to 12.39)	AT	.	1.61 (-3.67 to 6.88)
17.80 (4.78 to 30.81)	12.98 (0.50 to 25.45)	11.88 (-0.23 to 23.99)	9.87 (-2.82 to 22.56)	9.02 (-3.60 to 21.65)	8.70 (-3.84 to 21.25)	8.58 (-5.05 to 22.22)	8.38 (-5.32 to 22.09)	8.12 (-3.67 to 19.91)	7.54 (-6.18 to 21.27)	6.10 (-8.24 to 20.44)	4.80 (-2.64 to 12.24)	2.58 (-10.02 to 15.18)	NES	.
16.82 (10.63 to 23.02)	12.00 (7.04 to 16.97)	10.91 (6.96 to 14.85)	8.90 (1.95 to 15.85)	8.05 (2.73 to 13.37)	7.73 (2.14 to 13.32)	7.61 (0.05 to 15.17)	7.41 (-0.28 to 15.10)	7.14 (2.01 to 12.27)	6.57 (-1.00 to 14.14)	5.13 (-9.91 to 20.16)	3.83 (-4.87 to 12.53)	1.61 (-3.67 to 6.88)	-0.97 (-12.42 to 10.47)	Control

Outcome are provided as odds ratios (OR) and 95% CI. Odds ratios lower than 1 favor the column-defining intervention for the network meta-

analysis results (lower triangle) and the row-defining intervention for the pair-wise meta-analysis results (upper triangle)

Abbreviation: AM: aromatherapy massage; AT: acupoint therapy; C: control, CD: cool dialysate; DA: dopamine agonist; G: gabapentin; IAE: intradialytic aerobic exercise; ISE: intradialytic stretching exercise; NES: neuromuscular electrical stimulation; Vit-C: vitamin C; Vit-E: vitamin E; R: reflexology

Supplemental Table 6 Rank probability matrix of component NMA

Rank	1	2	3	4	5	6	7	8	9	10	11	12	13
Acupoint therapy	0.000	0.000	0.000	0.001	0.003	0.009	0.015	0.035	0.102	0.252	0.217	0.220	0.146
Aromatherapy message	0.025	0.175	0.300	0.208	0.133	0.072	0.043	0.029	0.010	0.004	0.001	0.000	0.000
Control	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.007	0.071	0.294	0.391	0.237
Cool dialysate	0.861	0.095	0.020	0.014	0.006	0.003	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Dopamine agonist	0.000	0.003	0.029	0.076	0.102	0.175	0.215	0.189	0.125	0.064	0.018	0.004	0.000
Gabapentin	0.030	0.09	0.140	0.130	0.151	0.118	0.119	0.101	0.061	0.035	0.011	0.004	0.001
Intradialytic aerobic exercise	0.000	0.008	0.014	0.025	0.026	0.042	0.043	0.067	0.130	0.236	0.175	0.204	0.030
Intradialytic stretching exercise	0.054	0.506	0.219	0.119	0.053	0.020	0.017	0.011	0.0010	0.000	0.000	0.000	0.000
Iron	0.012	0.032	0.070	0.073	0.095	0.093	0.091	0.117	0.171	0.118	0.074	0.037	0.017
Neuromuscular electrical stimulation	0.001	0.001	0.002	0.005	0.012	0.011	0.014	0.016	0.036	0.077	0.141	0.117	0.567
Reflexology	0.011	0.048	0.130	0.196	0.174	0.144	0.097	0.106	0.071	0.017	0.005	0.001	0.000
Vitamin C	0.000	0.013	0.035	0.067	0.134	0.200	0.191	0.165	0.123	0.047	0.020	0.005	0.000
Vitamin E	0.006	0.020	0.041	0.086	0.111	0.113	0.154	0.164	0.163	0.079	0.044	0.017	0.002

Supplemental Table 7. Assessment of confidence of evidence of network meta-analysis of RLS symptom relief

Comparison	Within-study bias	Reporting bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence
Acupoint therapy:Control	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
Aromatherapy massage:Control	No concerns	Undetected	No concerns	No concerns	No concerns	No concerns	Low
Control:Cool dialysate	No concerns	Undetected	No concerns	No concerns	No concerns	No concerns	High
Control:DA	Major concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Very low
Control:Intradialytic aerobic exercise	Major concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Very low
Control:Intradialytic stretching exercise	Major concerns	Undetected	No concerns	No concerns	No concerns	No concerns	Low
Control:Iron	Major concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Very low
Control:Reflexology	Major concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Very low
Control:Vitamin C	No concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
Control:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
Control:Vitamin E	No concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Moderate
Cool dialysate:Intradialytic stretching exercise	No concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Moderate
DA:Gabapentin	No concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Moderate
DA:Intradialytic aerobic exercise	Major concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Very low
DA:Vitamin C	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
Intradialytic aerobic exercise:Intradialytic aerobic exercise+DA	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
Intradialytic aerobic exercise:Neuromuscular electrical stimulation	Major concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Very low

Intradialytic stretching exercise:Reflexology	Major concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
Vitamin C:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
Vitamin C:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
Vitamin C+Vitamin E:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low
Acupoint therapy:Aromatherapy massage	No concerns	Undetected	No concerns	No concerns	Some concerns	Major concerns	Low
Acupoint therapy:Cool dialysate	No concerns	Undetected	No concerns	No concerns	No concerns	Major concerns	Low
Acupoint therapy:DA	Major concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Very low
Acupoint therapy:Gabapentin	Major concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Very low
Acupoint therapy:Intradialytic aerobic exercise	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Acupoint therapy:Intradialytic aerobic exercise+DA	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Acupoint therapy:Intradialytic stretching exercise	Major concerns	Undetected	No concerns	No concerns	Some concerns	Major concerns	Very low
Acupoint therapy:Iron	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Acupoint therapy:Neuromuscular electrical stimulation	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Acupoint therapy:Reflexology	Major concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Low
Acupoint therapy:Vitamin C	No concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Low
Acupoint therapy:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Acupoint therapy:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Aromatherapy massage:Cool dialysate	No concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Low
Aromatherapy massage:DA	No concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Low
Aromatherapy massage:Gabapentin	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low

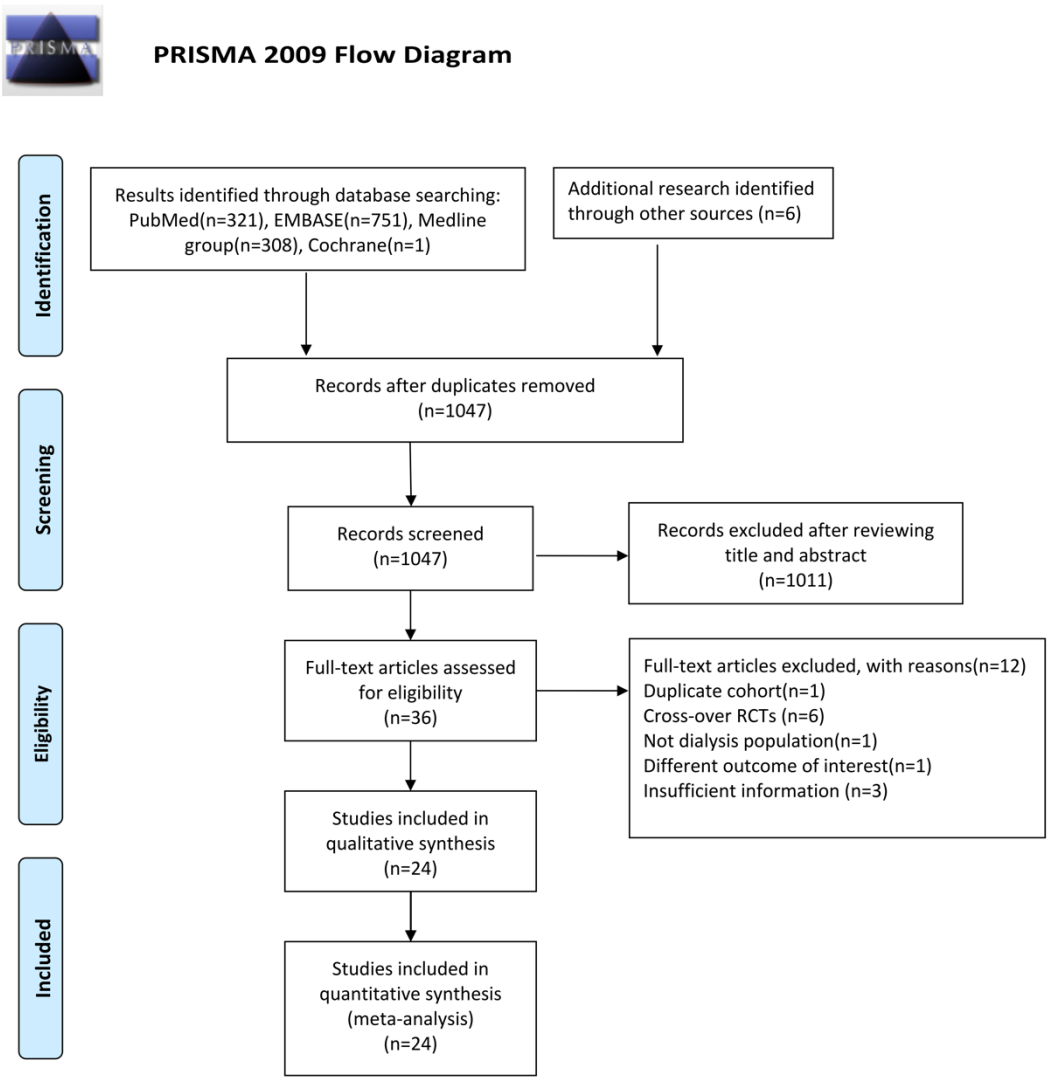
Aromatherapy massage:Intradialytic aerobic exercise	Major concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Very low
Aromatherapy massage:Intradialytic aerobic exercise+DA	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Aromatherapy massage:Intradialytic stretching exercise	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Aromatherapy massage:Iron	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Aromatherapy massage:Neuromuscular electrical stimulation	Major concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Very low
Aromatherapy massage:Reflexology	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Aromatherapy massage:Vitamin C	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Aromatherapy massage:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Aromatherapy massage:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Control:Gabapentin	Major concerns	Undetected	No concerns	No concerns	Some concerns	Major concerns	Very low
Control:Intradialytic aerobic exercise+DA	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Control:Neuromuscular electrical stimulation	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Cool dialysate:DA	No concerns	Undetected	No concerns	No concerns	Some concerns	Major concerns	Low
Cool dialysate:Gabapentin	No concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Low
Cool dialysate:Intradialytic aerobic exercise	Major concerns	Undetected	No concerns	No concerns	Some concerns	Major concerns	Very low
Cool dialysate:Intradialytic aerobic exercise+DA	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Cool dialysate:Iron	Major concerns	Undetected	No concerns	No concerns	Major concerns	Major concerns	Very low

Cool dialysate:Neuromuscular electrical stimulation	Major concerns	Undetected	No concerns	No concerns	No concerns	Major concerns	Very low
Cool dialysate:Reflexology	No concerns	Undetected	No concerns	No concerns	Major concerns	Major concerns	Low
Cool dialysate:Vitamin C	No concerns	Undetected	No concerns	No concerns	Major concerns	Major concerns	Low
Cool dialysate:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Low
Cool dialysate:Vitamin E	No concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Low
DA:Intradialytic aerobic exercise+DA	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
DA:Intradialytic stretching exercise	Major concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Very low
DA:Iron	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
DA:Neuromuscular electrical stimulation	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
DA:Reflexology	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
DA:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
DA:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Gabapentin:Intradialytic aerobic exercise	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Gabapentin:Intradialytic aerobic exercise+DA	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Gabapentin:Intradialytic stretching exercise	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Gabapentin:Iron	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Gabapentin:Neuromuscular electrical stimulation	Major concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Very low
Gabapentin:Reflexology	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Gabapentin:Vitamin C	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Gabapentin:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Gabapentin:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low

Intradialytic aerobic exercise:Intradialytic stretching exercise	Major concerns	Undetected	No concerns	Some concerns	Some concerns	Major concerns	Very low
Intradialytic aerobic exercise:Iron	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Intradialytic aerobic exercise:Reflexology	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Intradialytic aerobic exercise:Vitamin C	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Intradialytic aerobic exercise:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Intradialytic aerobic exercise:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Intradialytic aerobic exercise+DA:Intradialytic stretching exercise	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Intradialytic aerobic exercise+DA:Iron	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Intradialytic aerobic exercise+DA:Neuromuscular electrical stimulation	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Intradialytic aerobic exercise+DA:Reflexology	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Intradialytic aerobic exercise+DA:Vitamin C	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Intradialytic aerobic exercise+DA:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Intradialytic aerobic exercise+DA:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Intradialytic stretching exercise:Iron	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Intradialytic stretching exercise:Neuromuscular electrical stimulation	Major concerns	Undetected	No concerns	No concerns	Major concerns	Major concerns	Very low
Intradialytic stretching exercise:Vitamin C	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low

Intradialytic stretching exercise:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Intradialytic stretching exercise:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Iron:Neuromuscular electrical stimulation	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Iron:Reflexology	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Iron:Vitamin C	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Iron:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Iron:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Neuromuscular electrical stimulation:Reflexology	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Neuromuscular electrical stimulation:Vitamin C	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Neuromuscular electrical stimulation:Vitamin C+Vitamin E	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Neuromuscular electrical stimulation:Vitamin E	Major concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Very low
Reflexology:Vitamin C	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Reflexology:Vitamin C+Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low
Reflexology:Vitamin E	No concerns	Undetected	No concerns	Major concerns	No concerns	Major concerns	Low

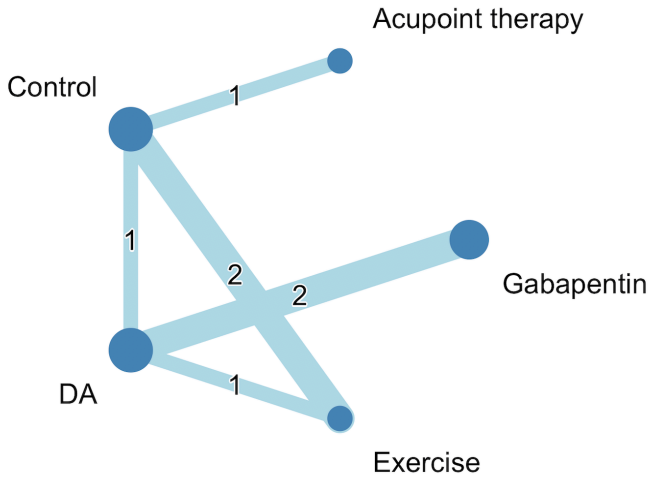
Supplemental Figure 1. PRISMA Flow Diagram



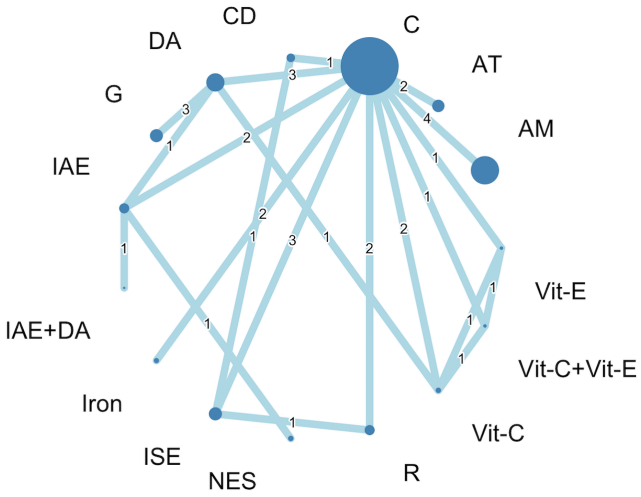
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For more information, visit www.prisma-statement.org.

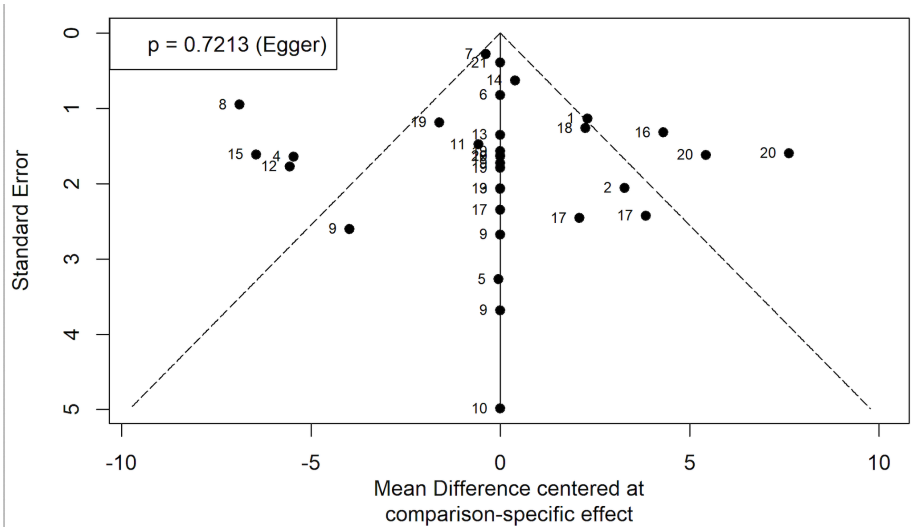
Supplemental Figure 2A.



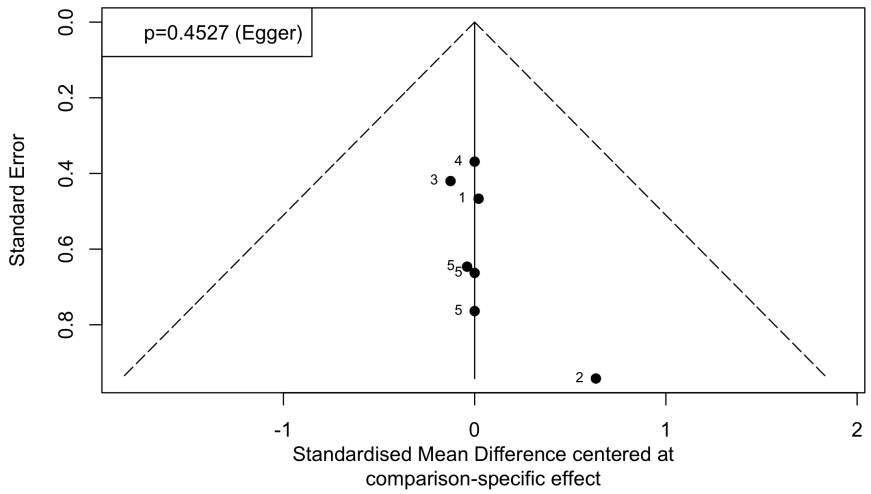
Supplemental Figure 2B



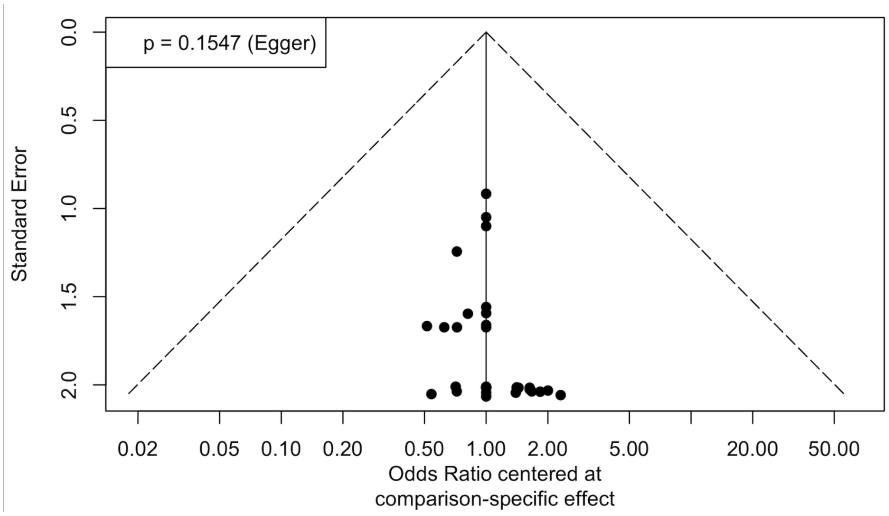
Supplemental Figure 3A.



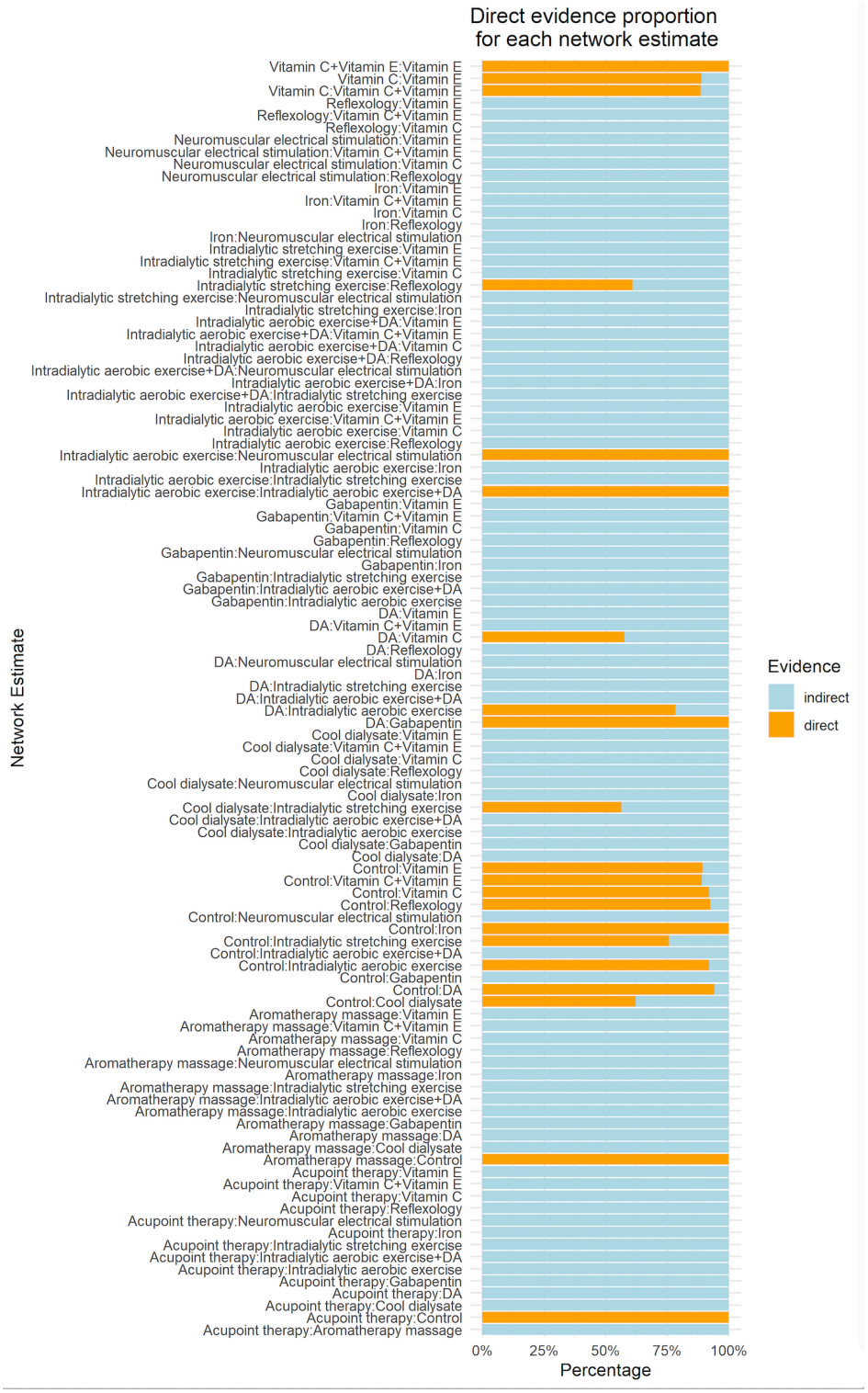
Supplemental Figure 3B.



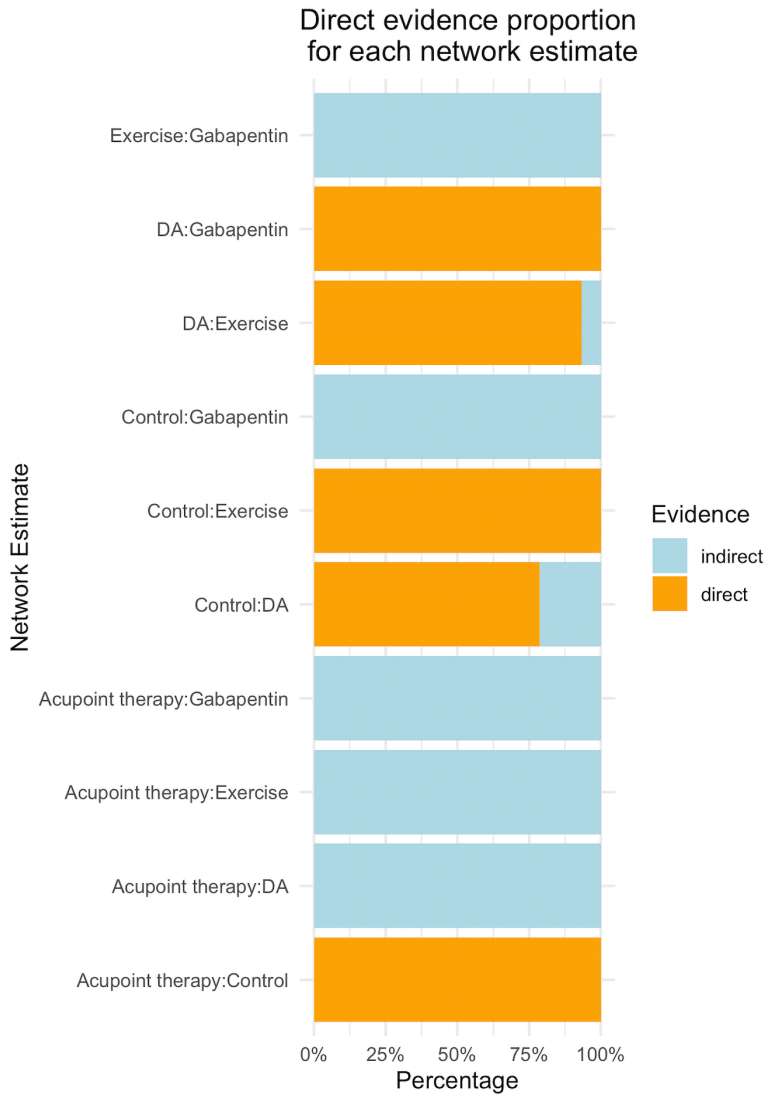
Supplemental Figure 3C.

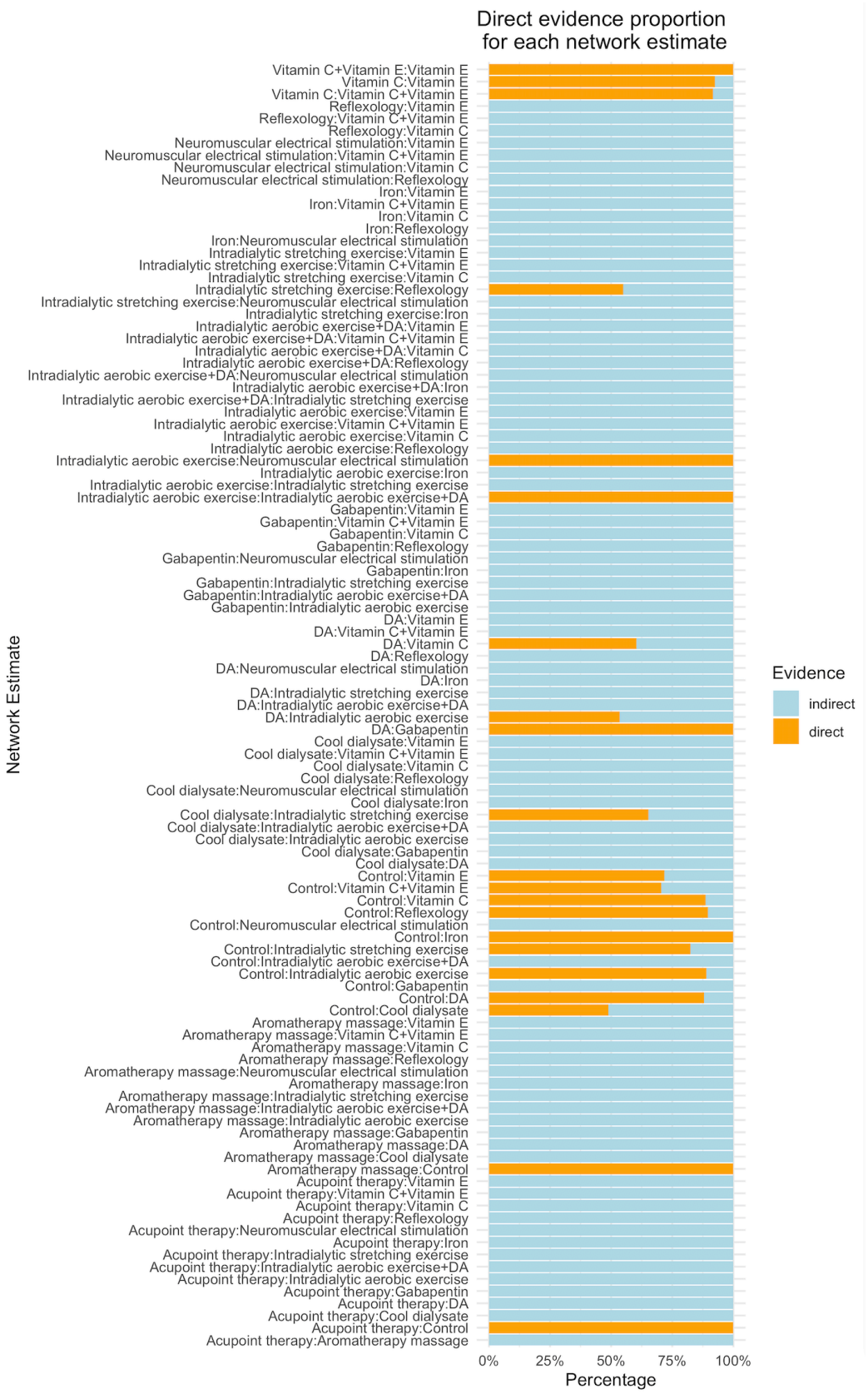


Supplemental Figure 4A.



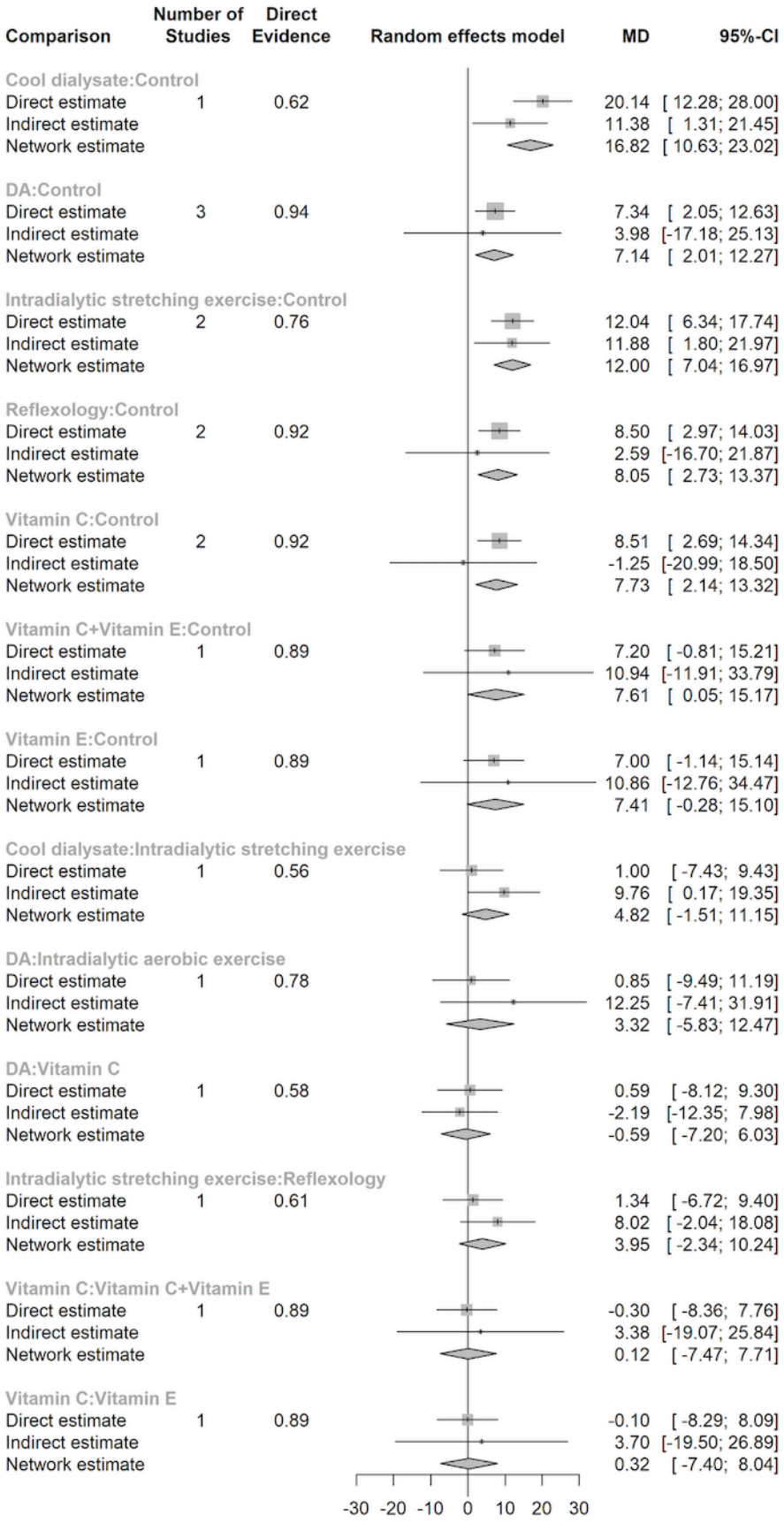
Supplemental Figure 4B.



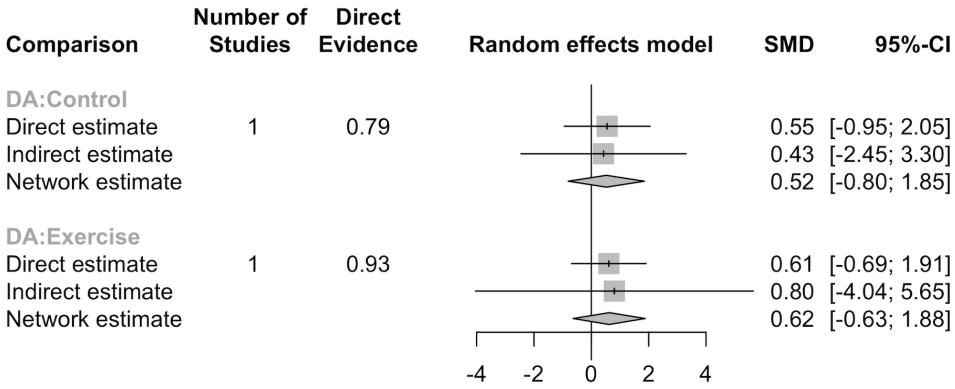


Supplemental Figure 4C.

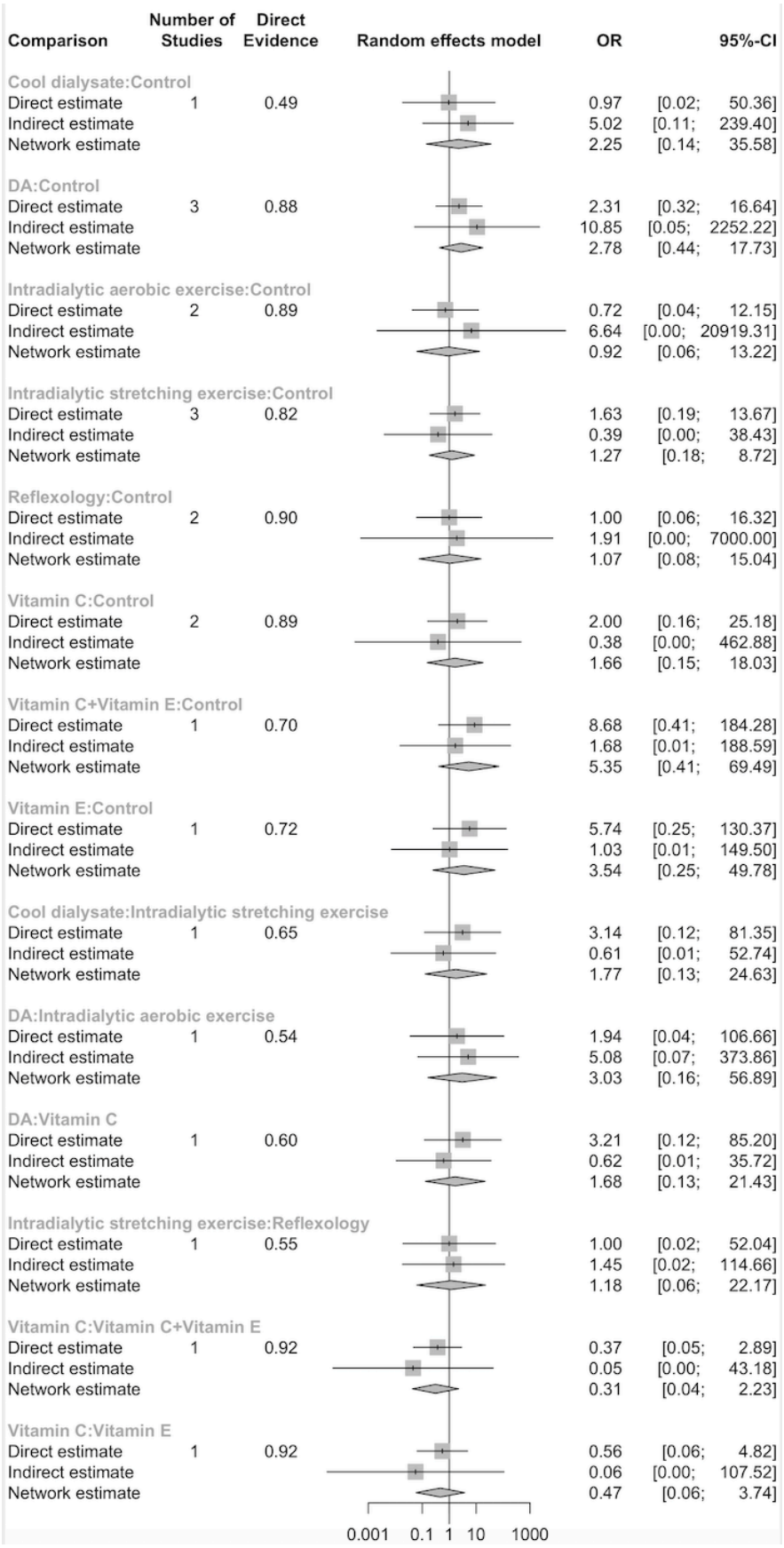
Supplemental Figure 5A.



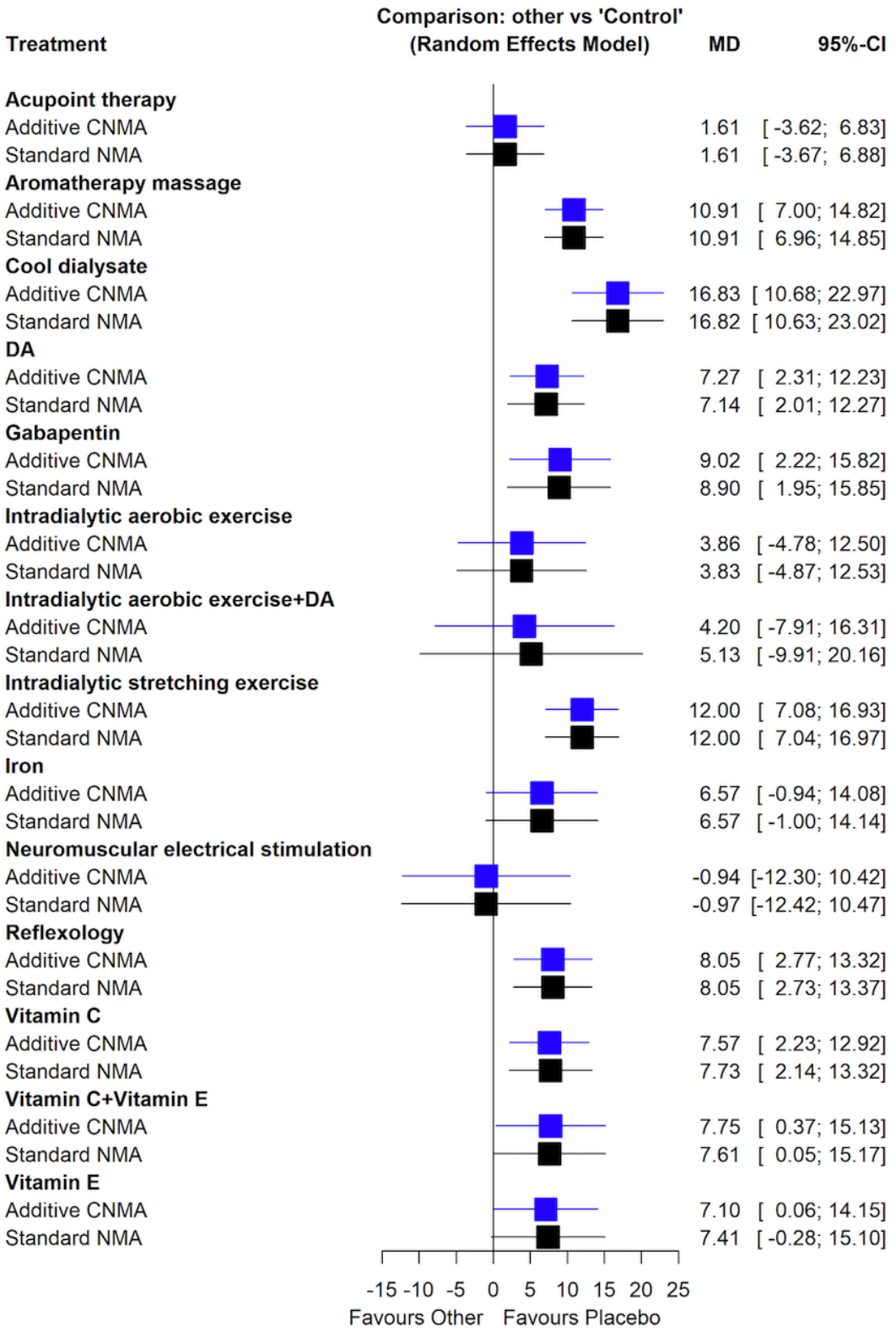
Supplemental Figure 5B.



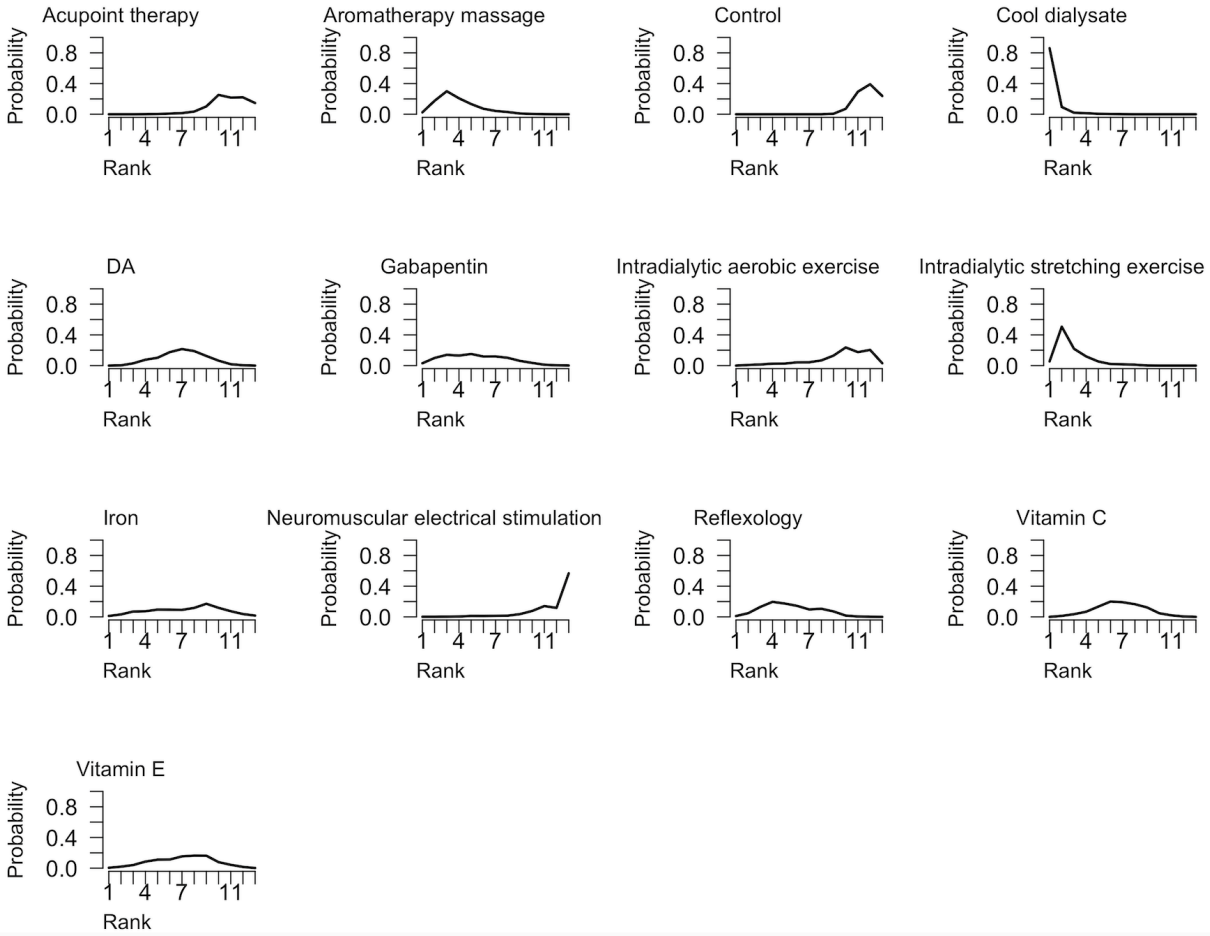
Supplemental Figure 5C.



Supplemental Figure 6.



Supplemental Figure 7. Results of computing ranking probabilities for each intervention components for RLS symptom relief



Supplemental Figure 8. Assessment of risk of bias of including studies.

